USER INTERFACE DESIGN

EXPERIMENT 10

AIM:

To create data visualizations such as Pie Chart and Bar Graph for an Inventory Management System using HTML, CSS, JavaScript, and Chart.js.

PROCEDURE:

Step 1: Creating the HTML Structure

Created an HTML page containing:

1. Heading – Inventory Stock Dashboard
2. Canvas for Pie Chart
3. Canvas for Bar Graph
4. Included Chart.js library for generating charts.

Step 2: Styling the Page using CSS

Applied CSS styling to improve the appearance of the dashboard:

- Light gradient background
- Center aligned heading
- Styled chart containers
- Rounded corners
- Box shadow effect
- Responsive layout

Step 3: Adding JavaScript Data

Defined inventory stock data for different categories:

- Mobiles – 180
- Fashion – 130
- Groceries – 220
- Furniture – 90
- Books – 140

Step 4: Creating Pie Chart

Used Chart.js to generate a Pie Chart showing stock distribution of different product categories.

Step 5: Creating Bar Graph

Used Chart.js to generate a Bar Graph showing stock quantity available in each category.

CODE:

INDEX.HTML:

```
<!DOCTYPE html>

<html lang="en">

<head>

<meta charset="UTF-8">

<meta name="viewport" content="width=device-width, initial-scale=1.0">



<title>Inventory Dashboard</title>



<script
src="https://cdnjs.cloudflare.com/ajax/libs/Chart.js/4.4.1/chart.umd.js"></script>



<style>

body{

font-family: Arial, sans-serif;

text-align:center;

background: linear-gradient(to right,#dbeafe,#eff6ff);

margin:0;

padding:20px;

}
```

```
h1{
color:#1e3a8a;
margin-bottom:30px;
}

.container{
display:flex;
justify-content:center;
gap:30px;
flex-wrap:wrap;
}

.chart-box{
width:500px;
height:400px;
background:white;
padding:20px;
border-radius:25px;
box-shadow:0 5px 15px rgba(0,0,0,0.1);
```

```
}

canvas{
width:100% !important;
height:100% !important;
}
</style>

</head>
<body>

<h1>Inventory Stock Dashboard</h1>

<div class="container">

<div class="chart-box">
<canvas id="pieChart"></canvas>
</div>

<div class="chart-box">
```

```
<canvas id="barChart"></canvas>

</div>


</div>


<script src="style.js"></script>


</body>

</html>
```

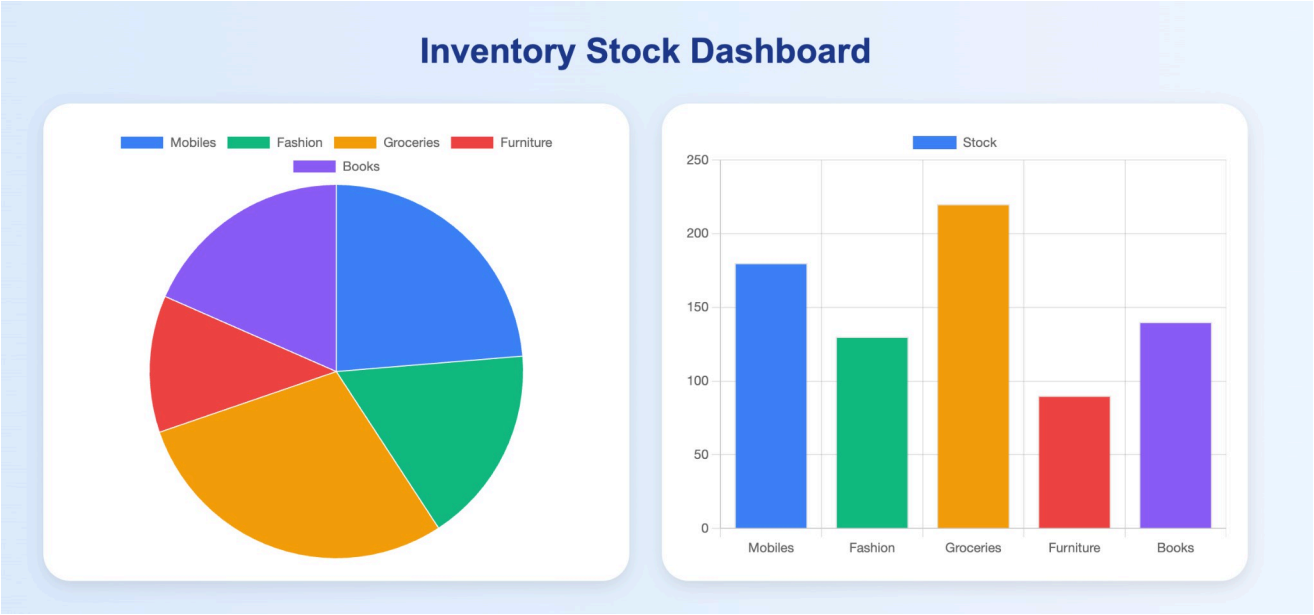
STYLE.JS:

```
const labels = ['Mobiles','Fashion','Groceries','Furniture','Books'];


const data = {

labels: labels,

datasets: [{

label:'Stock',

data:[180,130,220,90,140],

backgroundColor:[

'#3b82f6',

'#10b981',
```

```
'#f59e0b',
'#ef4444',
'#8b5cf6'
],
borderWidth:1
}]
};


// Pie Chart
new Chart(document.getElementById('pieChart'), {
type:'pie',
data:data,
options:{
responsive:true,
maintainAspectRatio:false
}
});


// Bar Chart
new Chart(document.getElementById('barChart'), {
```

```
type:'bar',

data:data,

options:{

responsive:true,

maintainAspectRatio:false,

scales:{

y:{

beginAtZero:true

}
}

}

});
```

OUTPUT:



The Inventory Stock Dashboard was created successfully.

● Pie Chart displays stock distribution among product categories.

● Bar Graph displays quantity of stock available in each category.

● Attractive and responsive dashboard interface was created successfully.

RESULT:

Successfully created data visualizations for an Inventory Management System using JavaScript and Chart.js. The Pie Chart and Bar Graph clearly represent stock distribution and available quantity in an effective manner.